

Extended-Koopmans -Theorem Approach to *Ab Initio* Calculations upon the Ground State and First Excited State of the LiH Anion

LUDWIK ADAMOWICZ

Quantum Theory Project, University of Florida, Gainesville, Florida 32611

JAMES C. ELLENBOGEN

MITRE Corporation, McLean, Virginia 22101

E. A. McCULLOUGH, JR.

Department of Chemistry and Biochemistry, Utah State University, Logan, Utah 84322

1. Introduction

This paper explores the extended-Koopman's theorem (EKT) [1,2] approach to the old, but difficult problem of determining the energies of attachment of an extra electron to a neutral molecule. The numerical multiconfiguration Hartree-Fock (MC-HF) method for diatomics [3] was applied to neutral LiH to generate $V^>$, the EKT one-particle effective potential for electron attachment to the neutral molecule. Separate MC-HF calculations were performed upon the LiH anion to explore the constrained EKT wavefunctions for the system.

The theoretical problem of electron attachment to a neutral molecule is particularly intriguing when that N -electron system has a large, permanent dipole moment μ . When μ exceeds 1.625 Debye, an elementary quantum-mechanical model reveals that, within the Born-Oppenheimer approximation, there should be an infinite number of bound states for the $(N + 1)$ -th electron [4,5]. Detailed knowledge of a number of these so-called "dipole" states would be necessary for perturbation-theory calculations of the dynamical behavior of the molecule anion.

Thus, an *ab initio* tool which, with a single fast calculation, yields the energies and wavefunctions for a number of these dipole states all at once is worthy of further investigation. The extended Koopman's theorem is such a tool. It describes a one-particle method for obtaining variational approximations to the electron affinities (EA) and $(N + 1)$ -particle *anionic* states from a many-particle calculation upon only the *neutral* ground state. This method was set forth by Day, Smith, and Garrod [1] and by Smith and Day [2] more than a decade ago; but, until now, no computational tests of the technique have appeared in the literature.

2. Theory

In the extended-Koopman's approach it is assumed that one already knows the N -electron, *multi-configuration* ground-state eigenfunction

$$\Psi_0^{(N)}(x_1, \dots, x_N) = \sum_J^R C_J \Phi_J^{(N)}(x_1, \dots, x_N) \quad (1)$$

of the many-particle (model) Hamiltonian $\hat{H}$. The orthonormal configuration functions $\Phi_J^{(N)}$ are expressed as linear combinations of Slater determinants constructed from the set of orthonormal spin orbitals $\{\phi_i(x_i) | i = 1, \dots, M\}$, where $M \geq N$. The constants C_J are the configuration coefficients. Then, an *ansatz* for the k -th $(N + 1)$ -electron *anionic* state is given by

$$\Psi_k^{(N+1)}(x_1, \dots, x_N, x_{N+1}) = \hat{A}[u_k^>(x_{N+1})\Psi_0^{(N)}(x_1, \dots, x_N)] \quad (2)$$

where the $\hat{A}$ is an antisymmetrizing operator, and the one-particle function for the $(N + 1)$ -th electron is expressed as the linear combination of spin orbitals

$$u_k^>(x_{N+1}) = \sum_i^M c_{ik}^>\phi_i(x_{N+1}) \quad (3)$$

In general, each $u_k^>$ is nonorthogonal to all ϕ_i of the same spatial symmetry which are occupied in the MC-HF wavefunction (1). However, all the $u_k^>$ will be orthogonal to any ϕ_p which is doubly occupied in *every* determinant of $\Psi_0^{(N)}$, because, in that case, $c_{pk}^>$ vanishes.

The essence of the extended-Koopman's idea is that a *variational* approximation to the exact $(N + 1)$ -electron anionic states and electron-attachment energies may be determined by varying only (the linear parameters $c_{ik}^>$ in) the one-particle-function $u_k^>(x_{N+1})$ to minimize the difference in energy

$$\varepsilon_k^> = \frac{\langle \Psi_k^{(N+1)} | \hat{H} | \Psi_k^{(N+1)} \rangle}{\langle \Psi_k^{(N+1)} | \Psi_k^{(N+1)} \rangle} - \langle \Psi_0^{(N)} | \hat{H} | \Psi_0^{(N)} \rangle \quad (4)$$

between the $(N + 1)$ -electron function $\Psi_k^{(N+1)}$ and that of the fixed neutral "reference" state $\Psi_0^{(N)}$. The formal minimization of this difference produces a set of one-particle matrix equations [2]

$$V^>C^> = S^>C^>\varepsilon^> \quad (5)$$

which may be solved to determine through Eq. (3) a set of M optimal one-electron functions $\{u_k^>(x_{N+1})\}$ and a set of M variationally optimal approximate electron affinities, $\{EA_k\}$:

$$EA_k = -\varepsilon_k^> \quad (6)$$

An unconstrained minimization of the difference in Eq. (4) would always lead to the ground-state solution for the anion. Solutions for higher anionic states arise from the imposition of appropriate orthogonality constraints.

Equation (14) of Ref. 1 defines the one-particle effective potential matrix $V^>$ in terms of a matrix $V^<$, which may be identified with the Fock operator of MC-HF theory [6]. The matrix or "overlap" matrix $S^>$ is defined

$$S^> = 1 - \gamma \quad (7)$$

in terms of the representatives γ_{ij} of the first-order density matrix [7]

$$\begin{aligned}\gamma(x_1, x'_1) &= \int dx_2 \dots \int dx_N \Psi_0^{(N)}(x_1, x_2, \dots, x_N) \Psi_0^{(N)*}(x'_1, x_2, \dots, x_N) \\ &= \sum_i^M \sum_j^M \phi_i(x_1) \gamma_{ij} \phi_j^*(x'_1)\end{aligned}\quad (8)$$

in the space $\{\phi_i\}$.

In the special case that the neutral reference state consists of only a variationally optimal (Hartree-Fock) single configuration $\Psi_0^{(N)}$, the extended-Koopman's approach reduces to the ordinary Koopman's approach, the orbitals $u_k^>$ are merely the Hartree-Fock canonical *virtual* orbitals, and the eigenvalues $\varepsilon_k^>$ are the corresponding virtual orbital energies.

It has been demonstrated numerous times [8] that the canonical virtual orbital energies from single-configuration Hartree-Fock calculations are poor approximations to the exact or experimental electron affinities. This may be explained by the fact that a change in the electron correlation energy, which is very significant in the addition or removal of the $(N + 1)$ -th electron [9], is poorly represented by the difference between the energy expectation values of two single configurations. (The difference between the total energies given by separate many-electron Hartree-Fock calculations on neutral and anionic states also produces poor agreement with experimental EAs.)

However, in the extended-Koopman's approach both the neutral state $\Psi_0^{(N)}$ and the anionic state $\Psi_k^{(N+1)}$ are represented by multiconfiguration wavefunctions, as may be seen by inspection of Eq. (1) and by substituting Eqs. (1) and (3) into Eq. (2) to expand the EKT anionic wavefunction:

$$\begin{aligned}\Psi_k^{(N+1)}(x_1, \dots, x_N, x_{N+1}) &= \sum_J^R \sum_i^M C_J c_{ik}^> \hat{A}[\phi_i(x_{N+1}) \Phi_J^{(N)}(x_1, \dots, x_N)] \\ &= \sum_J^R \sum_i^M C_J c_{ik}^> \Phi_{Ji}^{(N+1)}(x_1, \dots, x_N, x_{N+1})\end{aligned}\quad (9)$$

The variational freedom is constrained in the configuration space of the extended-Koopman's multiconfiguration wavefunction for the anion; but it permits some change in the electron correlation from that represented in the neutral state. Calculations to be described in the subsequent sections of this paper investigate the extent to which such a constrained anionic wavefunction is successful in representing the change in correlation energy arising from the extra electron in the dipole states of LiH^- .

3. Computational Method

The *numerical* MC-HF method for diatomics, which we have described in detail elsewhere [3], was used to calculate wavefunctions for the neutral ground state of LiH in three different approximations that will be designated by I, II, and III:

$$\text{I: } \Psi_0^{(4)} = C_1 |1\sigma\overline{1\sigma}2\sigma\overline{2\sigma}| + C_2 |1\sigma\overline{1\sigma}3\sigma\overline{3\sigma}| \quad (10a)$$

$$\begin{aligned} \text{II: } \Psi_0^{(4)} = C_1 |1\sigma\overline{1\sigma}2\sigma\overline{2\sigma}| + C_2 |1\sigma\overline{1\sigma}3\sigma\overline{3\sigma}| \\ + C_3 (|1\sigma\overline{1\sigma}1\pi_+\overline{1\pi_-}| + |1\sigma\overline{1\sigma}1\pi_-\overline{1\pi_+}|) \end{aligned} \quad (10b)$$

$$\begin{aligned} \text{III: } \Psi_0^{(4)} = C_1 |1\sigma\overline{1\sigma}2\sigma\overline{2\sigma}| + C_2 |1\sigma\overline{1\sigma}3\sigma\overline{3\sigma}| + C_3 |1\sigma\overline{1\sigma}4\sigma\overline{4\sigma}| \\ + C_4 (|1\sigma\overline{1\sigma}1\pi_+\overline{1\pi_-}| + |1\sigma\overline{1\sigma}1\pi_-\overline{1\pi_+}|) \\ + C_5 (|1\sigma\overline{1\sigma}1\delta_+\overline{1\delta_-}| + |1\sigma\overline{1\sigma}1\delta_-\overline{1\delta_+}|) \end{aligned} \quad (10c)$$

The overbar designates β spin orbitals.

Corresponding to these MC-HF wavefunctions, which account for electron correlation only in the 2σ shell, the following total energies were obtained for the neutral LiH molecule: I: $E^{(4)} = -8.003735$ a.u., II: $E^{(4)} = -8.016043$ a.u., and III: $E^{(4)} = -8.021725$ a.u., respectively.

For each of the three neutral-state calculations, the lowest five-electron anionic state of Σ symmetry was approximated by a constrained MC-HF procedure. The optimal sets of numerical orbitals $\{\phi_i\}$ and configuration coefficients $\{C_K\}$ produced for the neutral four-electron ground state of LiH by these three calculations were used to construct three wavefunctions for the five-electron anion according to Eq. (2); i.e.

$$\Psi^{(5)}(x_1, \dots, x_4, x_5) = \hat{A}[u(x_5)\Psi_0^{(4)}(x_1, \dots, x_4)] \quad (11)$$

where the function u is of type σ .

The function u need only be orthogonal to the 1σ orbital which is double-occupied in every configuration. Thus, this function for the MC-HF wavefunctions I, II, and III can be expressed as follows:

$$\text{I: } u = c_2 2\sigma + c_3 3\sigma + \phi^\perp \quad (12a)$$

$$\text{II: } u = c_2 2\sigma + c_3 3\sigma + \phi^\perp \quad (12b)$$

$$\text{III: } u = c_2 2\sigma + c_3 3\sigma + c_4 4\sigma + \phi^\perp \quad (12c)$$

where $\phi^\perp$ signifies a function orthogonal to all the others used in the equation.

To obtain the optimal functions u , three numerical MC-HF calculations for the LiH anion were performed in which only the coefficients c and the orbital functions $\phi^\perp$ of Eqs. (12) were optimized to get the five-electron wavefunctions of Eqs. (11) and the corresponding total energies for the anions. The electron affinities were calculated in these three approximations by subtracting these energies one at a time from the total energy that was calculated for neutral LiH. The procedure outlined in this paragraph produces anionic wavefunctions and first electron affinities which are exactly equivalent to those which are produced by the extended Koopman's procedure of Refs. 1 and 2.

The constrained MC-HF strategy just described can also be applied to find wavefunctions and electron affinities corresponding to the first excited state of the LiH anion. To generate this anionic excited state from the neutral ground state I: $\psi^{(4)}$, for example, the LiH⁻ configuration list is generated from the equation

$$\psi^{(5)}(x_1, \dots, x_4, x_5) = \hat{A}[u'(x_5)\psi_0^{(4)}(x_1, \dots, x_4)] \quad (13)$$

where the expansion of the function u' is

$$u' = c'_2 2\sigma + c'_3 3\sigma + c'_\perp \phi^+ + \phi^{++} \tag{14}$$

and where ϕ^{++} is a function orthogonal to the orbitals 1σ , 2σ , 3σ , and ϕ^+ . A numerical MC-HF calculation was performed to optimize only the three coefficients c' and the function ϕ^{++} by varying the *second* root of the Hamiltonian matrix in the configuration space. All core molecular orbitals, the function ϕ^+ , and the configuration coefficients from the neutral state were held fixed.

4. Numerical Results

The electron affinities calculated for the LiH molecule are presented in Tables I and II. Table I contains results obtained by the subtraction of the energies of the

TABLE I. Electron affinities for the LiH molecule ($R_{\text{Li...H}} = 3.015$ a.u.) as calculated via constrained MC-HF procedure applied to the anion.

	HF	MC-HF ^a			Other results
		I	II	III	
Dipole moment [Debye]	6.00	5.68	5.75	5.86	
Ground state EA [a.u.]	78×10^{-4}	65×10^{-4}	67×10^{-4}	72×10^{-4}	110×10^{-4b}
Excited state EA [a.u.]	86×10^{-6}	55×10^{-6}	63×10^{-6}	72×10^{-6}	95×10^{-6c} 106×10^{-6d}

^aFor definition of MC-HF functions I, II, and III see Eq. (10).
^bCI results of Ref. 10.
^cNumerical MC-HF result of Ref. 11.
^dResult of Ref. 12 obtained with the Coupled Cluster method and numerical orbitals.

TABLE II. Extended Koopman's orbital energies in a.u. for LiH MC-HF ground state II: $\Psi^{(4)}$ described in Eq. (10b) of the text. Orbital energies $\epsilon_k^>$, which represent approximate electron affinities $(EA)_k$ were determined by the solution of Eq. (4) of the text; orbital energies $\epsilon_k^<$, which represent approximate ionization potential, $(IP)_k$, were determined by the solution of Eq. (12) in Ref. 6.

k	$\epsilon_k^> \simeq -EA_k$	$\epsilon_k^< \simeq -IP_k$
1σ	—	-2.44610
2σ	1.02433	-0.29056
3σ	0.04512	-0.83034
4σ	-6.79×10^{-3}	—
5σ	-0.06×10^{-3}	—
1π	0.47306	-1.97743
2π	0.63410	0.23512

MC-HF neutral and MC-HF anionic states that were determined according to the strategy set forth in the last section of this paper, while Table II contains those obtained from the solution of Eq. (5). A comparison of the electron affinity results in Tables I and II shows that the positive electron affinities from the two computational procedures agree to the limit of the accuracy of the EKT procedure (i.e., the extent to which the MC-HF Fock matrix was converged). This confirms that the constrained MC-HF calculations upon the anions did generate the EKT anionic states.

From Table I it is evident that the electron affinities strongly depend on the extent to which correlation among the 2σ electrons is accounted for by the LiH and LiH⁻ MC-HF wavefunction. This dependence can be understood in terms of the dipole moments which are calculated for these same states. As is shown in Table I, the largest dipole moment is calculated for the single-determinant Hartree-Fock state and the smallest is calculated for the two-determinant MC-HF state II. As would be expected from simple argument based upon the Coulomb law, values for the electron affinities closely follow the trend in the dipole moment, with the largest electron affinities associated with the state having the largest dipole moment. It is interesting to note that the results obtained from the single-determinant LiH state are the closest to the more accurate results given in previous work that used the CI method [10], the MC-HF method [11], and the coupled cluster method [12].

To verify the equivalence between the present restricted MC-HF procedure and the extended-Koopman's-theorem method for determination of electron affinities, the effective Hamiltonian matrix $V^>$ was constructed with numerical MC-HF orbitals for each of the three neutral state calculations. Because the MC-HF calculations were converged to four places after the decimal point in all the off-diagonal elements, $V^>$, which is dependent upon the MC-HF Fock matrix [1,6], was symmetric to at least that accuracy. Next, the EKT secular equations (5) were solved to give the appropriate electron affinities. Results obtained for the MC-HF function II are displayed in Table II. The accuracy of these calculations within the EKT formalism was limited to five significant decimal places by the extent of convergence of the neutral-state MC-HF calculations.

As shown in Table II, it was also possible to diagonalize the $V^<$ operator given in Eq. (20) of Ref. 1, as calculated for LiH, to determine approximate ionization potentials $-\epsilon_k^<$ for the LiH molecule.

5. Summary and Conclusions

Constrained MC-HF calculations were used to construct wavefunctions in the extended Koopman's approximation for the ground state and first-excited state of the LiH molecular anion. Electron affinities for these two states were calculated by direct subtraction of the neutral LiH MC-HF total energy from the MC-HF total energies of these respective approximations to the anionic states. It was confirmed through calculations that this procedure is exactly equivalent to the extended-Koopman's one-particle approximation.

Three levels of MC-HF approximation, with various amounts of 2σ -shell correlation, were used in these calculations. This demonstrated that the electron affinities are very sensitive to factors which affect the calculated dipole moment. However,

even for the best MC-HF function used in this work, a substantial difference was found between the EKT electron affinities and those determined from very accurate calculations that appeared in earlier work. It is likely that this difference arises from the failure of the extended Koopman's approximation to account for electron correlation between the "extra" electron in the LiH anion and those electrons in 2σ core-shell of the molecule. This effect has been shown [10,12] to be important for the ground state of the anion, and it makes a significant contribution in the excited state as well [11,12]. Also, the EKT method fails to account for "relaxation" of the 2σ electrons upon addition of the extra electron, an effect which is especially significant for the ground state of the anion.

The combination of the above observation leads to the conclusion that the extended Koopman's approach to calculating electron affinities incorporates systematic error which will make it inadequate in some cases. The extended Koopman's procedure can be expected to work much better for molecules, whose dipole moment is strongly overestimated by the Hartree-Fock approximation. A good example is the BeO molecule. For this case the correlated dipole moment differs from the Hartree-Fock one by as much as 1.0 Debye [8]. However, systematic errors are likely to be manifested if the neutral reference state for the calculation is anything less than a full-CI wavefunction.

Acknowledgment

This research was supported in part by the National Science Foundation.

Bibliography

- [1] O. W. Day, D. W. Smith, and C. Garrod, *Int. J. Quantum Chem. Symp.* **8**, 501 (1975).
- [2] D. W. Smith and O. W. Day, *J. Chem. Phys.* **62**, 113 (1975).
- [3] L. Adamowicz and E. A. McCullough, *J. Chem. Phys.* **75**, 2475 (1981).
- [4] O. H. Crawford, *Proc. Phys. Soc. London* **91**, 279 (1967).
- [5] O. H. Crawford, *Mol. Phys.* **20**, 585 (1971).
- [6] J. C. Ellenbogen, D. W. Smith, O. W. Day, and R. C. Morrison, *J. Chem. Phys.* **66**, 4795 (1977).
- [7] P. O. Löwdin, *Phys. Rev.* **97**, 1474, 1490, 1509 (1955).
- [8] Y. Yoshioka and K. D. Jordan, *J. Chem. Phys.* **73**, 5899 (1980).
- [9] for atoms: E. Clementi, and A. D. McLean, *Phys. Rev. A* **133**, 419 (1964); for molecules: see Ref. 8.
- [10] B. Liu, K. O-Ohata, and H. Kirby-Docken, *J. Chem. Phys.* **67**, 1853 (1977).
- [11] L. Adamowicz and E. A. McCullough, *J. Phys. Chem.* **88**, 2045 (1984).
- [12] L. Adamowicz and R. J. Bartlett, *J. Chem. Phys.* **83**, 6268 (1985).

Received April 10, 1986

Accepted for publication June 11, 1986